Rate-Adaptive Vector Quantization for Deep Joint Source-Channel Coding

Jiwan Seo, *Member, IEEE*, Jeongseok Ha, *Senior Member, IEEE*, and Joonhyuk Kang, *Member, IEEE*

***Abstract*—Traditional deep learning-based joint source-channel coding (DeepJSCC) frameworks typically require separate models for different code rates and channel conditions, and remain incompatible with practical digital communication systems due to their analog transmission nature. To address these limitations, we propose a unified Rate-Adaptive Quantization-based DeepJSCC (RAQJSCC) framework that enables channel-aware rate adaptation within a single model while maintaining compatibility with conventional digital communication systems. The proposed framework achieves rate adaptivity through a rate-adaptive quantization (RAQ) mechanism that enables flexible codebook adaptation within a single model, supporting variable-rate operation without retraining. To improve robustness under vector quantization-based digital transmission, we design a cosine similarity-based codebook reordering (CSCR) mechanism that aligns quantized representations with modulation schemes and reduces the impact of index errors. Furthermore, we incorporate an adaptive modulation and coding (AMC) unit that jointly selects modulation orders and effective codebook sizes based on channel conditions, enabling dynamic link adaptation across diverse SNR regimes. Experimental results demonstrate that the proposed RAQJSCC framework outperforms existing DeepJSCC and separation-based methods in terms of reconstruction quality over a wide range of SNRs, while effectively mitigating the cliff effect. These results highlight the potential of RAQJSCC as a practical and scalable solution for next-generation wireless communication systems.**

***Index Terms*—Deep joint source-channel coding, vector quantization, rate adaptation, adaptive modulation and coding.**

I. Introduction

With the advent of next-generation wireless communication and networks, efficient and adaptive image transmission is crucial in emerging applications such as robotics, autonomous systems, and immersive AR/VR platforms. Following Shannon's separation theorem [1], most modern communication systems have been designed with a modular architecture that optimizes source compression and error correction independently. For instance, source and channel coding are typically applied separately (e.g., JPEG [2] and BPG [3] for compression, and LDPC [4] for error correction). Although separation is asymptotically optimal, it can be suboptimal under practical finite-blocklength constraints or uncertain channel conditions.

Joint source-channel coding (JSCC) has thus emerged as a promising alternative by jointly minimizing end-to-end distortion across compression and channel protection [5], [6]. Prior work has explored JSCC from classical perspectives, including graphical-model-based optimization of code parameters [7], tight finite-blocklength bounds and non-asymptotic strategies [8], and quantizer designs tailored for noisy channels [9]. Despite years of research, practical traditional JSCC designs based on simplified probabilistic models have not succeeded in developing codes that outperform state-of-the-art separated baselines and are applicable to a wide range of source-channel pairs.

Recent advances in machine learning (ML) have led to rapid progress in JSCC design [10], particularly in the context of semantic and goal-oriented communications for future wireless networks [11], [12]. Neural network-based JSCC can reduce coding latency by jointly handling compression and error correction [13]. DeepJSCC has shown competitive performance across various scenarios, such as exploiting noisy feedback [14], achieving high perceptual quality [15], and extending to multimodal data [16]. Compared to conventional separation-based source and channel coding schemes, DeepJSCC offers robustness to channel variations and graceful degradation when channel conditions deteriorate, thereby mitigating the *cliff effect*.

Despite these advantages, the practical deployment of DeepJSCC is often constrained by its analog transmission methodology, which sends arbitrary complex-valued symbols under power constraints. This analog interface can be incompatible with conventional digital communication hardware that is optimized for discrete modulation schemes and fixed symbol constellations. To bridge this gap, recent studies have explored digital DeepJSCC by constraining channel inputs to discrete symbol sets. A growing body of work has investigated digital DeepJSCC using discrete latent indices over noisy channels, including variational autoencoder (VAE)-based discrete latents [17], [18] and related designs for digital channels [19], [20], [21]. More directly targeting digital PHY compatibility for DeepJSCC systems, DeepJSCC-Q [22] transmits channel symbols through a fixed-size constellation, and joint coding-modulation (JCM) [23] further aims to adapt to operating channel conditions while avoiding the non-differentiability of digital modulation.

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As a prominent approach to digital DeepJSCC, neural vector quantization (VQ) models [24] provide a natural way to ensure compatibility with digital communication systems by mapping continuous latent representations into discrete codebook indices [25]. Such discrete representations can be directly encoded into bits and transmitted via standard modulation schemes. However, VQ-based digital JSCC suffers from an inherent sensitivity to bit errors, where even a single bit flip during transmission may lead to the selection of a completely different codebook vector, resulting in significant degradation in reconstruction quality [26]. Furthermore, existing VQ models typically rely on a fixed codebook size, which tightly couples the learned representation with the transmission rate, thereby requiring multiple models to support different rates or channel conditions. Various approaches have been proposed to enable variable-rate transmission within a single model. In particular, prior works have primarily achieved rate adaptivity by modifying the quantization mechanism itself, for example through residual quantization [26], [27], product quantization [28], or multi-codebook structures [29]. While effective, these approaches rely on altering the quantization structure to support variable-rate operation, which may increase system complexity and limit the consistency of the learned representation across different rates. This naturally raises the question of whether rate adaptivity can be achieved in a more direct manner by simply controlling the codebook size itself, while preserving a consistent encoder-decoder mapping.

In this work, we develop a unified VQ-based digital DeepJSCC framework, termed Rate-Adaptive Quantization-based DeepJSCC (RAQJSCC), that enables rate-adaptive transmission within a single model. Rather than relying on external rate-control mechanisms, we leverage the structure of the vector-quantized representation itself to support variable-rate operation, allowing a single model to operate across multiple rates while maintaining consistency in the underlying discrete representation. The proposed framework builds upon a rate-adaptive quantization (RAQ) mechanism introduced in [30], which enables flexible rate control within a shared codebook structure without requiring retraining or multiple model instances. While the RAQ mechanism provides a flexible way to control the codebook size within a shared representation, it has primarily been studied in the context of compression-oriented applications, such as generative modeling. Its direct application to JSCC systems, particularly in physical-layer communication scenarios involving digital modulation and channel-aware transmission, remains largely unexplored. To improve compatibility with practical digital communication systems, we further design a cosine similarity-based codebook reordering (CSCR) mechanism that aligns quantized representations with digital modulation schemes, reducing the impact of index errors under noisy channels. In addition, we incorporate an adaptive transmission strategy based on an adaptive modulation and coding (AMC) control unit, which jointly selects the modulation order and effective codebook size according to channel conditions, enabling dynamic link adaptation across diverse SNR regimes.

The main contributions of this paper are summarized as follows:

- We propose RAQJSCC, a unified VQ-based digital DeepJSCC framework that enables rate-adaptive transmission within a single model, eliminating the need for multiple models across different rates and channel conditions.
- We develop a RAQ mechanism that enables flexible rate control by directly adjusting the codebook size within a shared discrete representation, allowing a single model to operate across multiple transmission rates while preserving a consistent encoder-decoder mapping.
- We introduce a CSCR mechanism that enhances robustness to index errors by aligning the structure of the codebook with digital modulation schemes.
- We design an AMC-based transmission strategy that jointly adapts modulation orders and effective codebook sizes according to channel conditions, enabling practical link adaptation in digital communication systems.
- We demonstrate that RAQJSCC consistently outperforms existing digital DeepJSCC and separation-based approaches across a wide range of channel conditions, while mitigating the cliff effect.

The remainder of this paper is organized as follows. Section II reviews related work on VQ-based digital semantic communication and rate-adaptive DeepJSCC. Section III introduces the system model and reviews the fundamentals of VQ-based digital communication. Section IV presents the proposed framework. Section V provides experimental results and analysis under various channel conditions. Finally, Section VI concludes the paper and outlines potential directions for future work.

II. Related Work

A. VQ-Based Digital Semantic Communication

Unlike the Shannon paradigm, which guarantees correct reception regardless of the meaning of transmitted packets, semantic communication focuses on how transmitted symbols convey goal-oriented meanings and how effectively the received meanings influence the desired outcomes. Recent work has focused on developing semantic communication systems based on machine learning for future wireless networks [31], [32]. However, to ensure compatibility with digital communication systems, semantic features should be converted into bits for transmission. Vector quantization (VQ) can convert semantic features into bits by mapping data to a finite codebook for efficient compression and transmission. Since the vector-quantized VAE (VQ-VAE) introduced in [24] achieved promising results in various ML applications, several studies have attempted to merge semantic communication systems with VQ. In [33], a robust end-to-end semantic communication framework via masked VQ using adversarial training was proposed to suppress semantic noise in both the transmitter and receiver. In [25], a separation-based digital semantic communication system was developed that converts the encoder's output into conventional symbols via conventional channel coding and modulation. In [26], the authors addressed two key challenges in VQ-based communication systems: *incompatibility*,

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arising from the structural gap between a large-index-range codebook and digital constellation modulation, and *mismatch*, due to misaligned local relationships between code indices and code vectors. They introduced a residual VQ mechanism to form a multi-level codebook that combines conventional VQ with digital modulation and employed Gray code-based codebook reordering to align nearby indices with similar vectors, thereby reducing channel noise susceptibility. Building on the hierarchical structure of residual VQ, recent work has also explored its potential for rate-adaptive transmission. For example, [27] explicitly leverages multi-stage VQ to enable flexible rate control by varying the number of active quantization stages. In [34], a GAN-based VQ model was constructed to map multi-channel indices using a single shared codebook, implementing OFDM modulation. In [35], the authors explored neural multivariate quantization for efficient transmission over cell-free massive MIMO systems. Building on these developments, recent advances have further extended VQ-based semantic communication frameworks toward video-domain scalability [36], attention-based channel adaptation conditioned on SNR [37], multi-codebook quantization design [29], and entropy-guided variable-rate quantization via selective sub-codebook activation [28].

B. Rate-Adaptive DeepJSCC

To maximize system efficiency and reliability, conventional communication systems dynamically adjust modulation schemes and code rates according to channel conditions through AMC. However, many existing DeepJSCC frameworks still require multiple trained models to operate across different channel SNRs and compression ratios, which increases deployment complexity. To address this limitation, various approaches have been proposed to improve adaptability within a single model. Early efforts explored bandwidth-agile transmission [38], flexible bandwidth allocation [39], attention-based compression ratio adjustment [40], and single-model multi-rate systems [41], [42], including approaches based on multi-stage residual VQ for flexible rate control [26], [27], highlighting the importance of reducing model redundancy while maintaining performance across diverse channel conditions.

More recent works have introduced more structured mechanisms for joint adaptation of rate and channel conditions. JCM [23] integrates source coding and digital modulation into a unified transmission system. In this context, [43] extends JCM to support multi-order modulation and entropy-guided rate adaptation via masking and modulation-conditioned submodules. In [28], the authors introduced an entropy-guided variable product quantization framework, where the transmission rate is adjusted by selectively activating sub-codebooks based on estimated entropy, effectively enabling dimension-level variable-rate coding. In [44], the authors proposed a transformer-based JSCC framework that adapts to varying SNRs and channel-use ratios via latent-channel masking, operating in an analog transmission regime. These works demonstrate that rate adaptation in DeepJSCC can be achieved through diverse mechanisms, including entropy-driven masking, sub-codebook activation, and modulation-aware feature

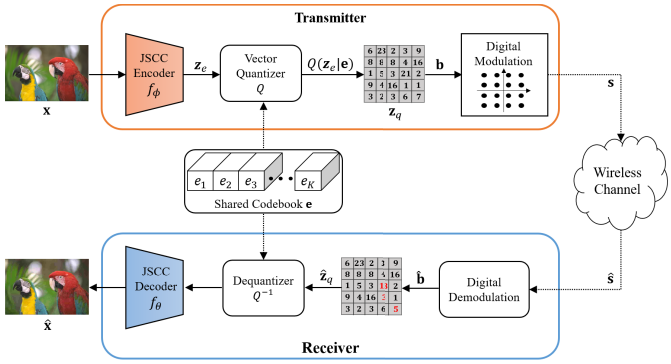


Fig. 1. System overview of the VQ-based DeepJSCC framework employing vector quantization and digital modulation for end-to-end wireless image transmission.

control, each offering different trade-offs in rate granularity, architectural complexity, and system compatibility. Within this line of research, the VQ-based approaches discussed above primarily achieve rate adaptivity by modifying the quantization mechanism itself, for example through product quantization [28], residual quantization [26], [27], or multi-codebook structures [29]. While these approaches provide flexible rate control, they typically rely on modifying the quantization structure itself, which may increase system complexity or affect the consistency of the learned discrete representation across different rates. In this work, we consider rate adaptivity within a purely VQ-based digital DeepJSCC framework by directly controlling the codebook size and its associated discrete representation, without modifying the underlying quantization structure. This perspective enables a single model to operate across multiple transmission rates while maintaining a consistent encoder-decoder mapping and compatibility with digital communication systems.

III. System Model

We consider an end-to-end wireless image transmission system in which a transmitter sends a source image $\mathbf{x}$ to a receiver over a noisy physical channel using a digital transceiver. As shown in Fig. 1, deep neural networks are leveraged for joint source-channel encoding and decoding while maintaining conventional digital modulation. Our goal is to design a practical digital JSCC framework that reconstructs the source image with low end-to-end distortion while adapting to varying link conditions.

A. Transmitter: Source-to-Channel Mapping via VQ

In the VQ-based DeepJSCC framework, the transmitter directly maps the source image to channel-compatible discrete representations without explicit separation between source and channel coding. Let $\mathbf{x} \in \mathbb{R}^{3\times H\times W}$ denote an RGB image drawn from dataset $\mathcal{D}$, where H and W are the image height and width. The transmitter first extracts a latent feature representation through an encoder network

$$\mathbf{z}_e(\mathbf{x}) = f_{\boldsymbol{\phi}}(\mathbf{x}), \tag{1}$$

where $f_{\boldsymbol{\phi}} : \mathbb{R}^{3\times H\times W} \to \mathbb{R}^{d\times h\times w}$ is parameterized by $\boldsymbol{\phi}$, and d, h, and w denote the channel dimension and spatial dimensions of the latent feature map, respectively.

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To enable digital transmission, each latent vector is quantized using a codebook $\mathbf{e}=\{e_i\}_{i=1}^{K}$ with $e_i \in \mathbb{R}^d$. For each spatial position (m,n), the quantizer Q selects the nearest codeword:

$$\mathbf{z}_q(\mathbf{x} \mid \mathbf{e})[m,n] = Q(\mathbf{z}_e(\mathbf{x})[m,n] \mid \mathbf{e}) = e_{c[m,n]}, \tag{2}$$

where

$$c[m,n] = \arg\min_{i\in\{1,\ldots,K\}} \|\mathbf{z}_e(\mathbf{x})[m,n] - e_i\|_2 . \tag{3}$$

This quantization step converts continuous features into discrete codeword indices, thereby defining the source representation as a sequence of indices. These indices determine the effective source coding rate in the digital JSCC system. Each quantized feature corresponds to a codebook index, forming

$$\mathbf{c} \in \{1,\ldots,K\}^{h\times w}. \tag{4}$$

Assuming $K = 2^k$, each index is represented by a k-bit binary label,

$$\mathbf{b}[m,n] = [b_1,\ldots,b_k] \in \{0,1\}^k. \tag{5}$$

These labels serve as inputs to digital modulation. Let M denote the modulation order, where the binary sequences are mapped to standard digital modulation schemes (e.g., M-QAM or M-PSK). Thus, the transmitter establishes a direct mapping from source images to discrete channel inputs through vector quantization.

The transmission rate is measured by the *bandwidth compression ratio* (BCR), defined as

$$\text{BCR} = \frac{N_{\text{sym}}}{3\cdot H\cdot W} \quad \text{channel symbols/pixel}, \tag{6}$$

where $N_{\text{sym}} = \frac{\log_2 K\cdot h\cdot w}{\log_2 M}$ denotes the total number of transmitted modulation symbols per input image. This interpretation is consistent with the *channel-use per pixel* in [45] and the *channel bandwidth ratio* in [26].

B. Physical Channel

The binary sequences $\mathbf{b}$ are mapped to a sequence of complex modulation symbols $\mathbf{s}$ according to the adopted modulation scheme and transmitted over a noisy channel. The received signal is expressed as

$$\hat{\mathbf{s}} = \mathbf{h}\odot\mathbf{s} + \mathbf{n}, \tag{7}$$

where $\mathbf{h}$ is the channel coefficient vector, and $\mathbf{n} \sim \mathcal{CN}(0,\sigma_n^2\mathbf{I})$ is additive white Gaussian noise (AWGN) with noise power σ_n^2. For AWGN channels, $\mathbf{h} = \mathbf{1}$. For Rayleigh fading, $\mathbf{h} \sim \mathcal{CN}(0,\mathbf{I})$. For Rician fading, $\mathbf{h} \sim \mathcal{CN}(\mu,\sigma_h^2\mathbf{I})$ with $\mu = \sqrt{K_{\text{factor}}/(K_{\text{factor}}+1)}$ and $\sigma_h = \sqrt{1/(K_{\text{factor}}+1)}$ where K_{factor} is the Rician K-factor. The received signal-to-noise ratio (SNR) is defined as $\bar{P}/\sigma_n^2$, where $\bar{P}$ denotes the normalized average transmit power.

C. Receiver: Reconstruction From Received VQ Indices

We assume perfect channel state information at the receiver, which is obtained via standard pilot-based channel estimation mechanisms as in conventional digital communication systems. After equalization, the receiver obtains

$$\tilde{\mathbf{s}} = \hat{\mathbf{s}} \oslash \mathbf{h}, \tag{8}$$

where $\oslash$ denotes element-wise division. The equalized symbols $\tilde{\mathbf{s}}$ are demodulated into reconstructed bits $\hat{\mathbf{b}}$, which are converted back into reconstructed indices $\hat{\mathbf{c}} \in \{1,2,\ldots,K\}^{h\times w}$. Using the shared codebook $\mathbf{e}$, the receiver forms the dequantized latent representation

$$\hat{\mathbf{z}}_q(\mathbf{x} \mid \mathbf{e}), \tag{9}$$

and reconstructs the image through a decoder network

$$\hat{\mathbf{x}} = f_{\boldsymbol{\theta}}\big(\hat{\mathbf{z}}_q(\mathbf{x} \mid \mathbf{e})\big), \tag{10}$$

where $f_{\boldsymbol{\theta}}: \mathbb{R}^{d\times h\times w} \to \mathbb{R}^{3\times H\times W}$ is parameterized by $\boldsymbol{\theta}$.

Channel errors manifest as perturbations in the transmitted indices, which lead to the selection of different codewords at the receiver. As a result, the reconstruction error is determined by the distance between the transmitted and received codewords. This implies that the end-to-end distortion is jointly governed by the source quantization and the channel-induced index errors.

D. Training Procedure

To train the VQ-based DeepJSCC system, we adopt a training strategy aligned with the VQ-VAE framework [24]. The encoder parameters $\boldsymbol{\phi}$, decoder parameters $\boldsymbol{\theta}$, and codebook embedding vectors $\mathbf{e}$ are jointly optimized to minimize a weighted pseudo-objective [46]:

$$\min_{f_{\boldsymbol{\phi}},f_{\boldsymbol{\theta}},\mathbf{e}} \mathop{\mathbb{E}}_{\mathbf{x}\in\mathcal{D}}\Big[\mathcal{L}_{\text{recon}}\big(f_{\boldsymbol{\theta}}(\hat{\mathbf{z}}_q(\mathbf{x}|\mathbf{e})),\mathbf{x}\big) + \alpha\mathcal{L}_{\text{cbk}}\big(\hat{\mathbf{z}}_q(\mathbf{x}|\mathbf{e}),\mathbf{z}_e(\mathbf{x})\big)\Big]. \tag{11}$$

Here, $\mathcal{L}_{\text{recon}}$ denotes the reconstruction loss between the source image $\mathbf{x}$ and the reconstructed output $\hat{\mathbf{x}}$, and $\mathcal{L}_{\text{cbk}}$ denotes the codebook loss that maintains consistency between the encoded features and their quantized features. The weighting parameter α balances the contributions of these two terms.

Since the quantization operation (2) and digital modulation are non-differentiable, we employ the straight-through estimator [47] for backpropagation. The overall loss can be written in the standard VQ form as

$$\mathcal{L}_{\text{VQ}}(\boldsymbol{\phi},\boldsymbol{\theta},\mathbf{e};\mathbf{x}) = \underbrace{\big\|\hat{\mathbf{x}}-\mathbf{x}\big\|_2^2}_{\mathcal{L}_{\text{recon}}} + \beta_1 \underbrace{\big\|\text{sg}\big[\mathbf{z}_e(\mathbf{x})\big]-\hat{\mathbf{z}}_q(\mathbf{x}|\mathbf{e})\big\|_2^2}_{\mathcal{L}_{\text{embed}}} + \beta_2 \underbrace{\big\|\text{sg}\big[\hat{\mathbf{z}}_q(\mathbf{x}|\mathbf{e})\big]-\mathbf{z}_e(\mathbf{x})\big\|_2^2}_{\mathcal{L}_{\text{commit}}}, \tag{12}$$

where sg[·] denotes the stop-gradient operator. The embedding loss $\mathcal{L}_{\text{embed}}$ updates the codebook vectors, while the commitment loss $\mathcal{L}_{\text{commit}}$ constrains the encoder outputs to remain close to the selected codewords. The codebook is initialized randomly and updated using exponential moving averages (EMA) during training [24]. The hyperparameters β_1 and β_2 control the relative contributions of the embedding and commitment losses, respectively, and are empirically set based on validation performance to balance reconstruction accuracy and codebook utilization.

To improve perceptual reconstruction quality, we replace the standard reconstruction loss with an enhanced objective incorporating LPIPS [48]:

$$\mathcal{L}'_{\text{recon}} = \|\hat{\mathbf{x}}-\mathbf{x}\|_2^2 + \lambda\mathcal{L}_{\text{LPIPS}}(\hat{\mathbf{x}},\mathbf{x}), \tag{13}$$

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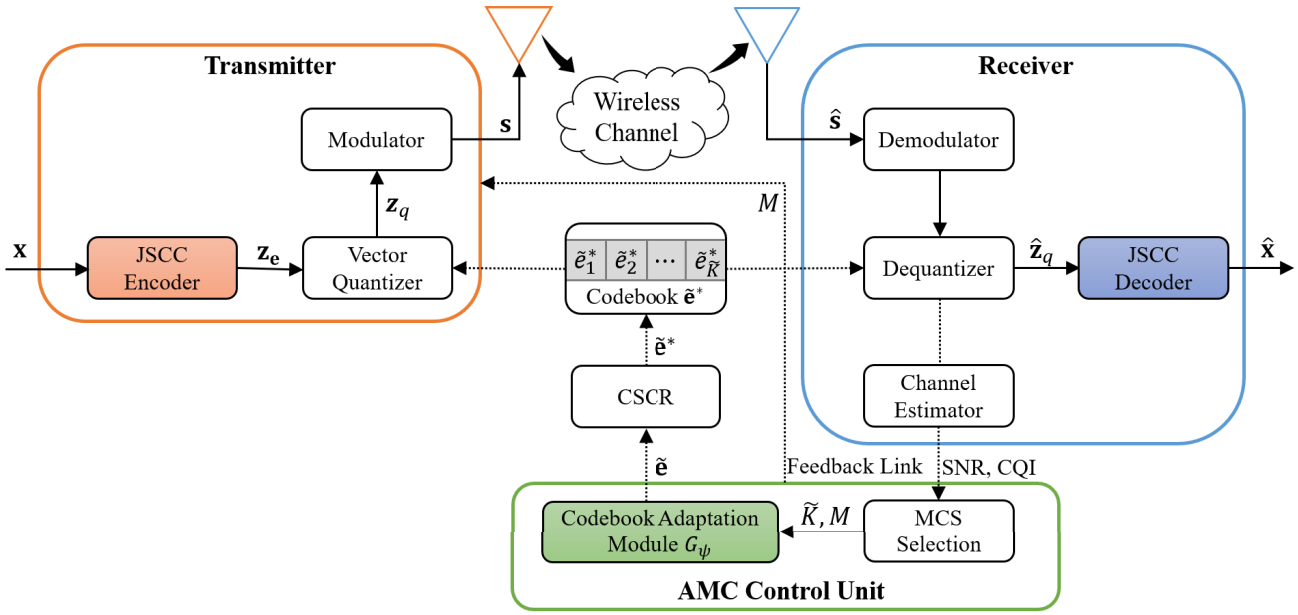


Fig. 2. Block diagram of the proposed RAQJSCC system, which integrates RAQ, CSCR, and AMC-based MCS selection for link adaptation.

where λ controls the contribution of the perceptual term, and LPIPS is computed using a pre-trained VGG network [49] (detailed formulation of $\mathcal{L}_{\text{LPIPS}}$ is presented in (31)).

The overall training objective jointly optimizes the encoder, decoder, and codebook to minimize reconstruction distortion under discrete representations. This distortion measure directly governs the robustness of the system to channel-induced index errors.

IV. Proposed RAQJSCC Framework

In this section, we present the proposed RAQJSCC framework, which extends the VQ-based digital DeepJSCC system described in Section III. As illustrated in Fig. 2, the proposed framework combines rate-adaptive quantization for flexible source-side representation, codebook reordering for improved robustness against index errors under digital transmission, and AMC-based transmission mode selection for link adaptation under varying channel conditions.

Conventional VQ-based digital transmission already provides a practical interface to standard modulation schemes through discrete codeword indices. However, several limitations remain when the system is operated over time-varying wireless links. A fixed codebook size inherently constrains the source representation to a single operating rate, the distortion caused by channel errors depends on how binary labels are assigned to codewords, and the modulation mode changes both the transmission reliability and the admissible symbol budget. These factors appear jointly in digital VQ transmission, as changes in the transmission mode simultaneously affect both the reliability of index delivery and the effective code rate determined by the codebook size.

We organize these aspects within a unified transmission structure. RAQ provides multiple adapted codebooks from a single base model, CSCR restructures the ordering of codewords to reduce the effect of index perturbations, and AMC selects the MCS mode according to channel conditions. The overall system maintains compatibility with practical digital transceivers while enabling operation across multiple codebook sizes and modulation orders within a single framework.

A. Rate-Adaptive Quantization

VQ-based DeepJSCC maps encoder outputs to discrete codeword indices, enabling compatibility with digital modulation schemes. In this structure, the codebook size K directly determines the effective code rate. As a result, a model trained with a fixed codebook size K operates at a *single* rate, and supporting multiple operating points typically requires training separate models.

To address this limitation, we introduce RAQ [30] into the digital DeepJSCC framework. Let $\mathbf{e} = \{e_i\}_{i=1}^{K}$ denote the anchor codebook. A codebook adaptation module G_ψ generates an adapted codebook

$$\widetilde{\mathbf{e}} = G_\psi(\mathbf{e}, \widetilde{K}) = \{\tilde{e}_i\}_{i=1}^{\widetilde{K}}, \tag{14}$$

where $\widetilde{K}$ is the target codebook size. The module is implemented as a sequence-to-sequence (Seq2Seq) model [50] that generates the adapted codebook in an autoregressive process, where each codeword $\tilde{e}_i$ is conditioned on previously generated codewords $\tilde{e}_{<i}$ and the anchor codebook $\mathbf{e}$. During training, the adaptation module G_ψ is optimized jointly with the encoder $f_{\boldsymbol{\phi}}$, decoder $f_{\boldsymbol{\theta}}$, and anchor codebook $\mathbf{e}$ to ensure consistency across different target sizes $\widetilde{K}$ via the standard VQ loss (12).

The role of RAQ in our framework is not merely to resize a codebook. Rather, it enables direct control of the code rate through codebook adaptation according to the channel-dependent operating point. This allows a single model to support multiple code rates without retraining separate encoder-decoder pairs. Given an adapted codebook $\widetilde{\mathbf{e}}$, the quantizer becomes

$$\widetilde{\mathbf{z}}_q(\mathbf{x} \,|\, \widetilde{\mathbf{e}}) = Q(\mathbf{z}_e(\mathbf{x}) \,|\, \widetilde{\mathbf{e}}), \tag{15}$$

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and the corresponding transmitted index matrix belongs to

$$\widetilde{\mathbf{c}} \in \{1, \ldots, \widetilde{K}\}^{h \times w}. \tag{16}$$

At deployment, the desired $\widetilde{K}$ is determined by the selected transmission mode, and the corresponding adapted codebook $\tilde{\mathbf{e}}$ is generated without retraining. This enables a single model to support multiple source rates while maintaining a shared embedding space. Unlike the previously proposed RAQ [30], our JSCC-oriented variant restricts $\widetilde{K}$ to values aligned with standard AMC constellation orders, and it selects the code rate online via real-time SNR feedback (see Section IV-C for further details). These modifications enable practical deployment within standard digital JSCC transceivers.

B. Codebook Reordering via Cosine Similarity

VQ-based JSCC systems are highly vulnerable to channel noise. A single bit error in a transmitted index can cause the receiver to select an entirely different embedding vector, resulting in a significant reduction in reconstruction quality. Existing DL-based quantization schemes [24] are prone to interference in fading scenarios because they are not designed to consider such conditions. In [26], the authors proposed a Euclidean distance-based codebook reordering to mitigate the index-vector mismatch. They also applied Gray coding to align with digital constellation design principles. A related study [36] employed cosine similarity to measure feature-level redundancy across adjacent frames for efficient index selection. While both approaches aim to improve robustness and efficiency, our method differs in that it structurally reorders the codebook itself based on angular similarity. We assign semantically similar vectors to adjacent binary indices in a systematic and comprehensive manner.

To enhance the efficiency of navigating high-dimensional codebook embedding vectors, we propose a *Cosine Similarity-based Codebook Reordering (CSCR)* strategy. Cosine similarity considers only the direction of vectors, making it insensitive to their magnitudes. This property is particularly advantageous in handling high-dimensional data, as it allows us to ignore differences in scale and focus on the orientation of embeddings. In high-dimensional spaces, the phenomenon known as the *curse of dimensionality* can render Euclidean distances less meaningful, but cosine similarity is less sensitive to this issue [51]. This motivates our use of the cosine similarity metric, as most large language models like BERT [52] evaluate the similarity between words or texts.

Our CSCR assigns similar binary code patterns to semantically close embedding vectors, thereby improving robustness to noise in the transmission channel. To begin CSCR, we compute the pairwise cosine similarity matrix $\mathbf{S}$ over all codebook embedding vectors $e_i, e_j \in \mathbb{R}^d$:

$$S_{i,j} = \frac{e_i^\top e_j}{\|e_i\| \, \|e_j\|} \quad \in [-1, 1]. \tag{17}$$

We then formulate the reordering problem as finding a permutation $\sigma \in \mathcal{P}_K$ that maximizes the cumulative cosine similarity between consecutive code vectors:

$$\max_{\sigma \in \mathcal{P}_K} \sum_{t=1}^{K-1} S_{\sigma(t), \sigma(t+1)}, \tag{18}$$

where $\mathcal{P}_K$ denotes the set of all permutations of $\{1, \ldots, K\}$. Since exhaustive search is infeasible for large K, we adopt a greedy approximation. Although this heuristic does not guarantee a global optimum, it efficiently ensures that locally similar code vectors become neighbors in the reordered codebook.

To provide a robust starting point, we select the codebook entry with the largest embedding norm:

$$i^{(1)} = \arg \max_{1 \le i \le K} \|\mathbf{e}_i\|. \tag{19}$$

A larger embedding norm reflects a more salient or dominant feature in the learned representation space, indicating that this vector carries significant semantic information [53], [54]. Empirically, initializing the greedy traversal at this high-norm vector anchors the sequence in a prominent region of the embedding space, which improves the overall effectiveness of the reordering heuristic.

Following this, the algorithm proceeds iteratively. At each iteration t, having chosen $i^{(t)}$, we remove it from the available set A_t and select the next vector as

$$i^{(t+1)} = \operatorname*{argmax}_{j \in A_t} S_{i^{(t)}, j}, \tag{20}$$

where A_t denotes the set of indices not yet selected. Finally, we map the selected indices to a Gray code sequence $\mathbf{G} = [g_1, \ldots, g_K]$ via a mapping function $\mathcal{M}$, producing the reordered codebook

$$\mathbf{e}^* = \{\mathbf{e}_{\mathcal{M}(g)} \mid g \in \mathbf{G}\}. \tag{21}$$

The role of Gray coding here is not merely conventional bit labeling. Combined with the semantic ordering induced by CSCR, it makes likely low-order bit perturbations correspond to nearby codewords in the reordered semantic space. Accordingly, CSCR acts as a source-channel interface design that reduces the semantic consequence of digital label errors, while preserving compatibility with standard digital modulation.

C. AMC Control Unit

Conventional AMC selects the modulation and coding scheme (MCS) based on channel quality to balance throughput and reliability. In contrast, the proposed AMC control unit (green block in Fig. 2) jointly determines both the modulation order and the effective code rate via RAQ, thereby controlling the end-to-end transmission mode of the digital DeepJSCC system.

Let the set of available MCS modes be

$$\mathrm{MCS}_l = (M_l, \widetilde{K}_l), \qquad l \in \mathcal{I} = \{0, \ldots, L-1\}, \tag{22}$$

where M_l is the modulation order (e.g., M_l-QAM constellation) and $\widetilde{K}_l$ is the adapted codebook size associated with mode l. To ensure a valid mapping from codebook indices to an integer number of modulation symbols, we require monotonic pairing between M_l and $\widetilde{K}_l$ as

$$\widetilde{K}_l = M_l^{\delta_l}, \qquad \delta_l \in \mathbb{N}, \tag{23}$$

so that the number of channel uses per source index

$$\delta_l = \frac{\log_2 \widetilde{K}_l}{\log_2 M_l} \tag{24}$$

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is an integer.[1] We design the set of MCS modes such that the modulation order is non-decreasing with the mode index (i.e., $M_l \le M_{l+1}$ for $l \in \{0, \ldots, L-2\}$). For modes sharing the same modulation order, the codebook size is strictly increasing, i.e., $\widetilde{K}_l < \widetilde{K}_{l+1}$ when $M_l = M_{l+1}$. With this design, each mode index l corresponds to a unique operating point with either a higher modulation order or a finer quantization level than lower-index modes. As a result, increasing l leads to improved spectral efficiency, quantization resolution, or both, enabling a consistent ordering of transmission modes.

Given the current post-equalization SNR γ, the receiver feeds back a channel quality indicator (CQI). Using an offline-estimated block error probability (BLEP) table $P_e(\gamma, l)$, the AMC control unit selects the highest-index feasible mode satisfying the reliability target:

$$l^\star = \max \left\{ l \in \mathcal{I} \;\middle|\; P_e(\gamma, l) \le P_{\text{trg}} \right\}, \tag{25}$$

where P_{trg} denotes the target BLEP.

Once $l^\star$ is selected, the codebook adaptation module G_ψ generates an adapted codebook of size $\widetilde{K}_{l^\star}$ as

$$\widetilde{\mathbf{e}}_{l^\star} = G_\psi(\mathbf{e}, \widetilde{K}_{l^\star}), \tag{26}$$

and is then reordered through CSCR:

$$\widetilde{\mathbf{e}}^*_{l^\star} = \text{CSCR}(\widetilde{\mathbf{e}}_{l^\star}). \tag{27}$$

The reordered codebook is shared by the transmitter and receiver for the current scheduling interval.

In practice, the AMC updates are triggered at regular scheduling intervals or when CQI crosses predefined thresholds, ensuring timely adaptation to channel variations. The end-to-end AMC pipeline at each scheduling interval can be summarized as:

$$\begin{aligned} \text{Receiver} Y \xrightarrow{\text{CQI}} \text{MCS Selection} \xrightarrow{(M_{l^\star}, \widetilde{K}_{l^\star})} G_\psi \\ \xrightarrow{\tilde{\mathbf{e}}_{l^\star}} \text{CSCR} \xrightarrow{\tilde{\mathbf{e}}^*_{l^\star}} \{\text{Quantizer, Modulator}\}. \end{aligned}$$

This adaptation mechanism allows the system to operate at the highest feasible code rate under the reliability constraint while mitigating the impact of channel-induced index errors.

D. Unconstrained Optimal MCS Selection

For completeness, we also consider an *unconstrained* MCS selection strategy that jointly selects the modulation order and codebook size to minimize the expected end-to-end distortion, without enforcing practical link-adaptation constraints.

Let $D(\gamma, M_l, \widetilde{K}_l)$ denote the end-to-end distortion under modulation order M_l and adapted codebook size $\widetilde{K}_l$, measured according to the reconstruction criterion defined in (13). The optimal mode is determined as

$$l^{\text{uncon}} = \arg\min_{l \in \mathcal{I}} \mathbb{E}\big[D(\gamma, M_l, \widetilde{K}_l)\big], \tag{28}$$

where the expectation is taken over the channel statistics.

In this setting, the modulation order and codebook size are selected solely to minimize distortion, without imposing reliability constraints such as BLEP thresholds or discrete MCS mapping. This unconstrained design serves as a performance upper bound on distortion-oriented mode selection, and is used for comparison with the practical AMC control unit.

[1] In our experiments, we considered $\delta_l \in \{1, 2, 3, 4\}$; in general, δ_l can be any positive integer to allow larger codebooks.

V. Experiments

A. Experimental Settings

We evaluate our approach using the ImageNet [55] and Kodak [56] datasets for general image reconstruction tasks. The ImageNet dataset comprises 1,281,167 training images, with approximately 1,000 images per class. For training our model and benchmark schemes, the images are resized to 256×256 pixels and center-cropped to 224×224 pixels. We perform the evaluation on the Kodak dataset, which offers 768×512 high-quality images commonly used for benchmarking image compression and reconstruction techniques. To enhance the robustness and generalization ability of the model, we apply random horizontal flipping as a data augmentation technique during ImageNet training.

B. Benchmarks

Unless otherwise specified, all schemes are evaluated over an AWGN channel. Additional experiments over Rician fading channels are also presented to evaluate the robustness of the proposed method under practical wireless propagation conditions. For ease of comparison, we express all BCRs as integer multiples of a common reference unit, denoted by ρ, where ρ is defined as the smallest BCR evaluated for the Kodak image resolution (768×512), corresponding to 0.02604 channel symbols per pixel. This notation provides a common rate scale for both the proposed method and all benchmark schemes (e.g., 1ρ, 2ρ, and 3ρ). All benchmarks are trained under noiseless conditions to ensure a fair comparison.

We integrate a VQ-VAE with a Seq2Seq model to enable RAQ, thereby enhancing the flexibility and performance of the encoder-decoder system. The Seq2Seq architecture comprises LSTM-based encoder and decoder modules along with embedding layers, following the approach in [30]. The JSCC encoder module employs a series of convolutional layers, including downsampling residual blocks and attention mechanisms, to extract and compress feature representations from the input images. The JSCC decoder mirrors the encoder's architecture, utilizing convolutional transpose layers for upsampling the feature maps back to the original image dimensions. Because the network is fully convolutional, doubling the input resolution simply doubles the codeword length, allowing us to test images of arbitrary size without architectural changes.

In our experiments, we consider the following approaches for performance evaluation:

- **RAQJSCC (proposed)**: We adopt a two-level hierarchical VQ-VAE [57] architecture, which serves as the foundational encoder-decoder framework for image reconstruction with $(\beta_1, \beta_2, \lambda) = (1, 0.25, 0.05)$. These hyperparameters follow standard practice in the VQ-VAE literature [24], [57] and have been shown to provide a good balance between codebook commitment and reconstruction fidelity. The proposed network contains 7.1 M parameters. During training, we uniformly sample

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$\widetilde{K} \in \{16, 17, \ldots, 4096\}$ to prevent overfitting to any single channel distribution. In AMC mode, the modulation order M is selected from $\{2, 4, 16, 64, 256\}$ (i.e., BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM) and the adapted codebook size $\widetilde{K}$ from $\{16, 64, 256, 1024, 4096\}$, yielding an effective BCR ranging from 1ρ to 4ρ. Average constellation power is normalized to $\bar{P} = 1$, and noise power is set by the target SNR.

- **DeepJSCC-Q** [22]: To ensure a fair comparison, we train several DeepJSCC-Q models (one for each combination of modulation order $M \in \{4, 16, 64, 256\}$ and BCR $k\rho$ with $k = 1, 2, 3$). Each network has 21.0 M parameters (approximately three times more than RAQJSCC's 7.1 M). The number of channel symbols is adjusted so that every model exactly meets its target BCR. The quantization strategy adopts soft-to-hard quantization for M-ary constellations. The majority of hyperparameters, such as the soft-to-hard hardness parameter, are set following [22]. Constellation power and noise scaling match the RAQJSCC setup. DeepJSCC-Q offers a fully specified digital-constrained JSCC and modulation protocol with sufficient implementation details in the literature, making it the most rigorous and reproducible JSCC baseline for our study.
- **JCM** [23]: The joint coding-modulation (JCM) framework is a digital semantic communication system that jointly learns source coding and probabilistic modulation mapping within a VAE. Instead of using fixed constellation points, JCM trains the probability distribution over discrete constellation symbols, effectively implementing a probability-based soft-decision modulation approach. To ensure a fair comparison, we follow the original configuration for modulation and training hyperparameters while employing the same encoder-decoder network as RAQJSCC. For consistency in training objectives, the reconstruction loss term in JCM is replaced with the same loss (13) used in RAQJSCC. The number of transmitted channel symbols is adjusted so that all models precisely satisfy the target modulation order and BCR (i.e., the combination of $M \in \{4, 16, 64\}$ and $k\rho$ with $k = 1, 2, 3$). Since the quantization module is omitted, JCM contains 5.7 M trainable parameters. While RAQJSCC employs a uniform classical constellation mapping, JCM learns its own adaptive constellation map, yielding a digital transmission model.
- **BPG + LDPC**: We employ a separation-based digital baseline with BPG [3] source coding and DVB-S2 LDPC channel coding. Each input image is first compressed using BPG at the closest possible compression quality corresponding to the target modulation and BCR setting. The resulting bitstream is then encoded with a DVB-S2 LDPC code (codeword length 16200 bits, rate 1/2) before modulation and transmission over the channel. While BPG provides state-of-the-art source compression performance, it is highly sensitive to bit errors in the entropy-coded stream. To mitigate the decoding failures, we apply an unequal error protection (UEP) mechanism that safeguards the first 64 header bytes against channel noise.

TABLE I
SUMMARY OF MCS CONFIGURATIONS: MODULATION SCHEME, BCR, AND MINIMUM REQUIRED SNR FOR EACH MCS INDEX

MCS Index	Modulation	BCR	Threshold SNR (dB)
0	BPSK	4ρ	2 dB
1	QPSK	1ρ	5 dB
2	QPSK	3ρ	9 dB
3	16-QAM	2ρ	11 dB
4	16-QAM	3ρ	15 dB
5	64-QAM	2ρ	18 dB
6	256-QAM	1ρ	29 dB

Both our RAQJSCC framework and the digital JSCC benchmarks, including DeepJSCC-Q and JCM, are designed to utilize digital modulation schemes, making them compatible with existing communication systems. This compatibility is crucial for practical deployment and integration into current communication systems. For statistical reliability, we train four independent instances of each model with different random seeds and report the results averaged over these runs. Throughout all experiments, SNR refers to E_b/N_0, where the effective bit rate accounts for both the selected modulation order and the corresponding codebook size.

C. Evaluation Metrics

To evaluate the reconstruction performance, we utilize the following metrics: (i) Peak Signal-to-Noise Ratio (PSNR); (ii) Multi-Scale Structural Similarity Index Measure (MS-SSIM); and (iii) Learned Perceptual Image Patch Similarity (LPIPS).

PSNR measures the ratio between the maximum possible power of a signal and the power of the corrupted noise affecting data fidelity. It is defined as

$$\mathrm{PSNR}(\mathbf{x}, \hat{\mathbf{x}}) = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}(\mathbf{x}, \hat{\mathbf{x}})}\right), \tag{29}$$

where $\mathrm{MSE}(\mathbf{x}, \hat{\mathbf{x}}) = \frac{1}{|\mathcal{D}|}\sum_{\mathbf{x}\in\mathcal{D}}(\mathbf{x} - \hat{\mathbf{x}})^2$ and MAX_I is the maximum possible pixel value of the image (e.g., $\mathrm{MAX}_I = 255$ for 8 bits per channel). Higher PSNR values indicate better pixel-wise fidelity and lower distortion between the original image $\mathbf{x}$ and the reconstructed image $\hat{\mathbf{x}}$.

MS-SSIM assesses structural similarity between two images by evaluating structural information over multiple scales [58]. It provides a more comprehensive assessment of image quality by extending the traditional SSIM to include image details at various resolutions. MS-SSIM is defined as

$$\mathrm{MS\text{-}SSIM}(\mathbf{x}, \hat{\mathbf{x}}) = \prod_{j=1}^{S}\left[\frac{(2\mu_{\mathbf{x}_j}\mu_{\hat{\mathbf{x}}_j} + C_1)(2\sigma_{\mathbf{x}_j\hat{\mathbf{x}}_j} + C_2)}{(\mu_{\mathbf{x}_j}^2 + \mu_{\hat{\mathbf{x}}_j}^2 + C_1)(\sigma_{\mathbf{x}_j}^2 + \sigma_{\hat{\mathbf{x}}_j}^2 + C_2)}\right]^{w_j}, \tag{30}$$

where S denotes the total number of scales. In (30), $\mu_{\mathbf{x}_j}$ and $\mu_{\hat{\mathbf{x}}_j}$ are the mean intensities of the original image $\mathbf{x}$ and the reconstructed image $\hat{\mathbf{x}}$ at scale j, respectively. Similarly, $\sigma_{\mathbf{x}_j}^2$ and $\sigma_{\hat{\mathbf{x}}_j}^2$ represent the variances, and $\sigma_{\mathbf{x}_j\hat{\mathbf{x}}_j}$ denotes the covariance between $\mathbf{x}$ and $\hat{\mathbf{x}}$ at scale j. The constants C_1 and C_2 are stabilization constants to prevent division by zero. The weights w_j are assigned to each scale, reflecting the relative importance of each scale in the overall similarity measure.

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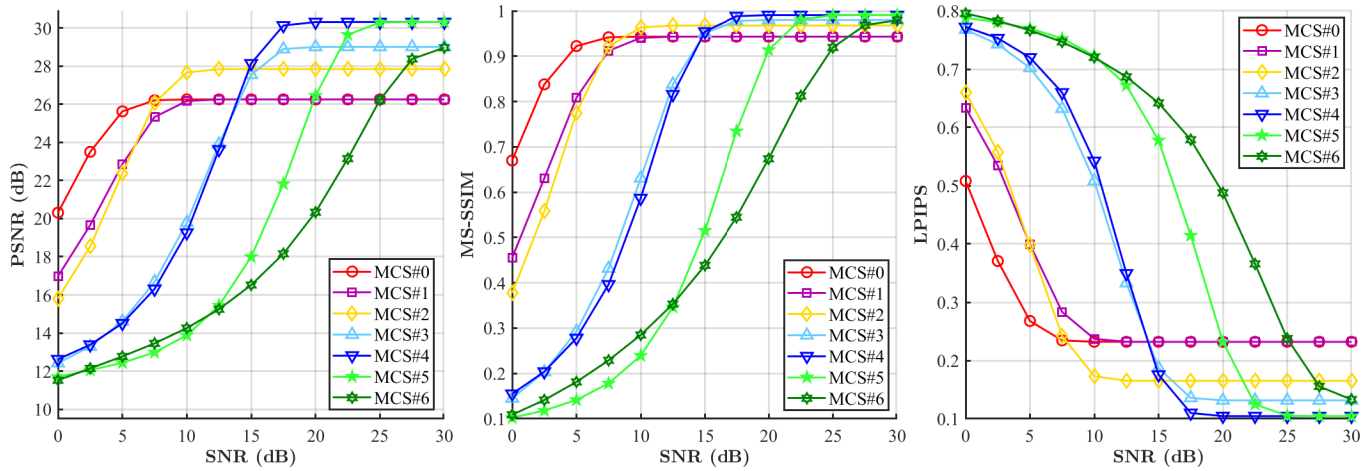


Fig. 3. RAQJSCC performance across different MCS indices under AWGN channels. Each curve corresponds to a modulation-BCR pair defined in Table I.

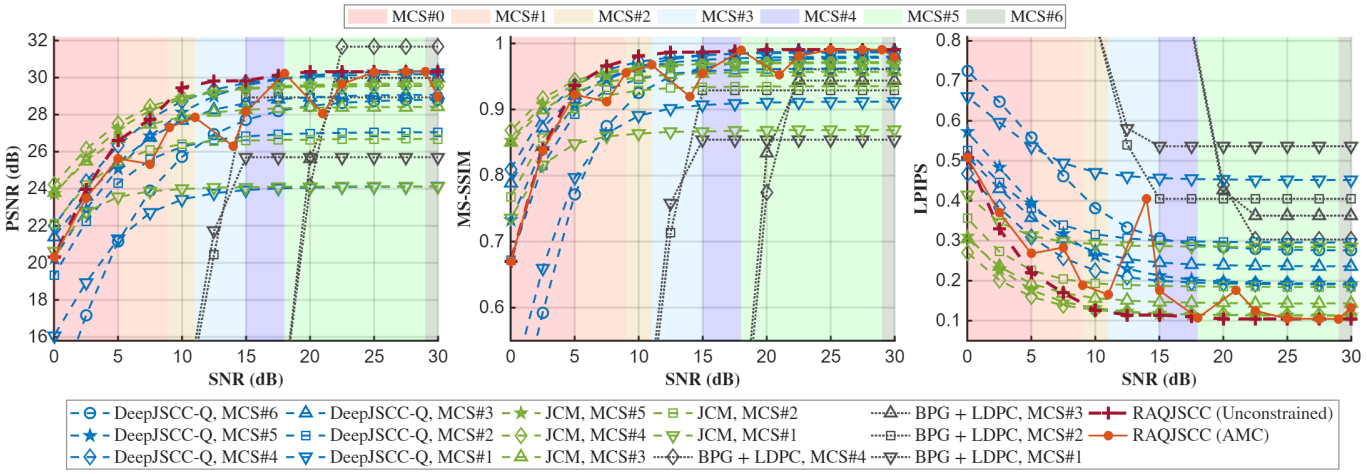


Fig. 4. Reconstruction performance comparison of RAQJSCC (AMC and Unconstrained), DeepJSCC-Q, JCM, and BPG+LDPC under AWGN channels. The evaluation is performed on the Kodak dataset using PSNR, MS-SSIM, and LPIPS as metrics.

Higher MS-SSIM values indicate better structural similarity and higher fidelity, as they imply lower levels of distortion across multiple scales.

LPIPS measures the perceptual similarity between two images by comparing deep feature representations extracted from a pre-trained neural network [48]. It is defined as

$$\mathcal{L}_{\text{LPIPS}}(\mathbf{x}, \hat{\mathbf{x}}) = \sum_{l=1}^{L} w_l \cdot \|\boldsymbol{\phi}_l(\mathbf{x}) - \boldsymbol{\phi}_l(\hat{\mathbf{x}})\|_2, \tag{31}$$

where $\boldsymbol{\phi}_l(\cdot)$ represents the feature maps obtained from the l-th layer of a pre-trained neural network. In (31), w_l are the learned weights that balance the contribution of the feature differences of each layer to the overall LPIPS score. L denotes the total number of layers from which features are extracted and compared. We use the VGG network [49] as the backbone network type. Lower LPIPS values indicate higher perceptual similarity and better reconstruction quality, as they signify smaller distances between the deep feature representations of the original and reconstructed images.

D. Performance Analysis

In this subsection, we first evaluate RAQJSCC's reconstruction performance under AMC for the seven MCS modes listed in Table I. We simplify the SNR thresholds for each MCS level by referencing the IEEE 802.11ac standard [59]. This is not intended to exactly match real 5G systems, which rely on CQI-based link adaptation under multipath fading and interference. Consequently, our MCS thresholds should be interpreted as a reference for AWGN-only simulations rather than a direct representation of practical 5G standards. Fig. 3 shows PSNR, MS-SSIM, and LPIPS for these MCS modes over an AWGN channel. At low SNR (below 10 dB), lower-order MCS indices (MCS #0 to MCS #2) dominate; as SNR increases, higher-order indices (MCS #4 to MCS #6) rapidly take over, achieving up to a 4 dB PSNR gain at high SNR (around 20 dB). Similar crossovers appear in MS-SSIM and LPIPS (center and right plots). Although we include only up to 256-QAM in our experiments, 1024-QAM or 4096-QAM could yield further gains at very high SNR. These results confirm that RAQJSCC effectively optimizes across all considered MCS indices.

Next, we compare a single RAQJSCC model against baselines trained at fixed modulation orders and BCRs. In Fig. 4, we plot PSNR, MS-SSIM, and LPIPS on the Kodak dataset for SNRs from 0 dB to 30 dB. We show both RAQJSCC (Unconstrained), which ignores AMC rules, and RAQJSCC (AMC), which follows the AMC schedule in Table I. As baselines,

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we include DeepJSCC-Q and JCM, two representative digital JSCCs that operate without an explicit demodulation stage. Although both DeepJSCC-Q and JCM discretize their transmitted symbols (DeepJSCC-Q through a soft-to-hard quantizer and JCM through a probabilistic modulator), they do not rely on an explicit demodulation step at the receiver. Instead, the receiver performs channel equalization and directly maps the continuous outputs to the reconstructed source signal.

The performance of the separation-based 'BPG+LDPC' schemes is shown by the black curves in Fig. 4. These schemes are configured at BCRs similar to our benchmarks. While these schemes achieve high reconstruction quality at sufficient SNRs, they exhibit the characteristic cliff effect at low SNR. DeepJSCC-Q, by contrast, degrades more gradually as SNR drops, consistent with prior studies [22]. In the low-SNR regime (0–5 dB), DeepJSCC-Q with 16-QAM (MCS #3 and #4) attains roughly 1.5 dB higher PSNR than RAQJSCC (though with a higher BCR and modulation order). We attribute this gap to their demodulation strategies. The JCM [23] exhibits a slightly different behavior: while its PSNR closely matches or marginally surpasses that of DeepJSCC-Q at SNRs above 10 dB, it achieves notably higher robustness in the low-SNR region, showing gains of 2–5 dB. This shows that JCM's probabilistic modulation provides smoother symbol transitions and an inherent degree of error tolerance. However, when compared with RAQJSCC under the same MCS configurations, JCM's advantage partly stems from adopting higher BCR and modulation orders even at low SNRs, thereby increasing its effective rate and compromising fairness in a direct comparison.

The difference originates from how the received symbols are processed. RAQJSCC employs a hard-decision demodulator that maps equalized symbols to discrete bit indices before decoding. Classical communication theory shows that hard-decision demodulation can incur an effective SNR penalty of 2–3 dB compared to soft-decision or continuous-valued processing [60]. On the other hand, DeepJSCC-Q and JCM retain soft-quantized symbol representations, allowing their neural decoders to exploit confidence levels in the received signal for more accurate reconstruction. This preservation of soft information underlies the graceful degradation property of DeepJSCC [12], [13], avoiding the abrupt performance cliffs typical of purely digital schemes. While explicit hard-decision demodulation in RAQJSCC can make it more vulnerable to noise at low SNRs, its AMC-based adaptability effectively compensates for this and enables seamless integration with existing communication hardware.

In terms of perceptual similarity (LPIPS), RAQJSCC achieves consistently stronger performance across the full SNR range. This primarily stems from the training objective: RAQJSCC is optimized using the LPIPS-based loss in (13), whereas DeepJSCC-Q is trained with an MS-SSIM objective. Consequently, DeepJSCC-Q attains excellent MS-SSIM scores but does not reach the same LPIPS performance, reflecting the inherent trade-off between pixel-level structural similarity and perceptual fidelity. To ensure a fair comparison with a modern digital semantic baseline, we apply the same LPIPS-based reconstruction loss (13) to JCM. Under this matched setting, JCM shows the strongest perceptual robustness, while RAQJSCC remains the best performer at moderate and high SNRs. This behavior suggests that the probabilistic modulation mechanism of JCM provides additional stability when noise dominates, but its advantage diminishes once the SNR becomes sufficiently high. However, this LPIPS-oriented robustness introduces a trade-off: JCM yields lower MS-SSIM scores than DeepJSCC-Q under identical MCS settings, indicating a reversed trend between perceptual quality and structural fidelity. For clarity, DeepJSCC-Q is trained using its original MS-SSIM objective in [22], which naturally leads to lower LPIPS but stronger structural similarity. Taken together, these results highlight how the choice of training objective fundamentally shapes which aspect of reconstruction quality (perceptual or structural similarity) a model emphasizes.

Another key contribution of RAQJSCC is its rate-adaptive quantization capability. Unlike non-adaptive approaches that require training separate models for each modulation and compression setting, RAQJSCC leverages a single model to operate across a wide range of conditions. This builds on the findings in [30], where a single RAQ-based model trained over multiple compression rates approached the performance of separately trained VQ-VAE models. Similarly, RAQJSCC dynamically adjusts its codebook and modulation configuration via the AMC control unit, adapting to real-time channel conditions without retraining. While RAQJSCC may show slightly lower performance at extremely low SNR compared to DeepJSCC-Q and JCM, which use soft-to-hard constellation mapping and continuous-valued features, it still exhibits graceful degradation, a characteristic common to DeepJSCC systems. This adaptability effectively mitigates the cliff effect commonly observed in separation-based schemes, avoiding abrupt performance collapse at low SNR. As a result, RAQJSCC maintains consistently robust reconstruction quality across diverse channel environments, distinguishing it from fixed-rate and non-adaptive designs.

We also observe that not all MCS selections provide optimal performance at intermediate SNR conditions. As shown in Fig. 3, when 'RAQJSCC (AMC)' uses MCS #6 (256-QAM, 1ρ), it yields lower PSNR and MS-SSIM values than certain benchmarks when the selected SNR is insufficient to fully support such a high-order modulation scheme. Moreover, at around 14 dB, the use of MCS #3 (16-QAM, 2ρ) results in a noticeable drop because the minimum SNR required for this MCS index is not met and the relatively higher code rate chosen cannot be effectively sustained. Although this MCS index aligns with the IEEE 802.11ac standard [59] for the given SNR range, our model-level analysis suggests that employing a lower modulation order or a higher code rate could have yielded better overall performance. These results suggest that while our adaptive mechanism provides flexibility, the MCS table and corresponding SNR thresholds may need to be refined for intermediate channel conditions. This argument is supported by the results of 'RAQJSCC (Unconstrained)', which selected the optimal MCS without constraints for each SNR. Increasing the granularity of the MCS indices or adjusting their required SNR levels could

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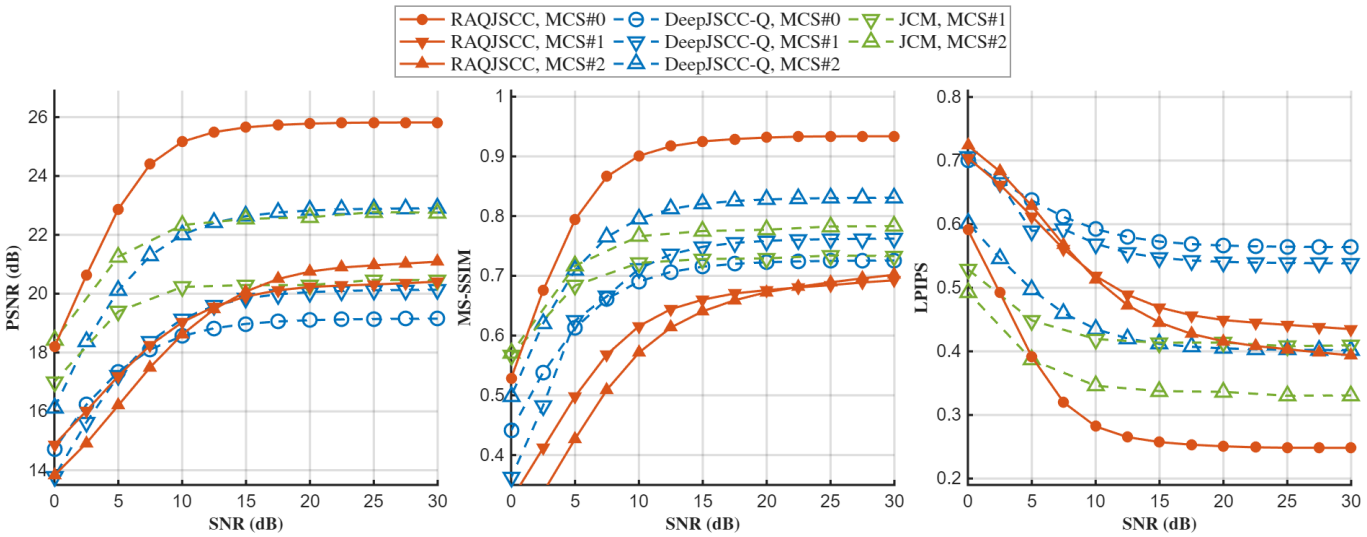


Fig. 5. Reconstruction performance comparison of RAQJSCC, DeepJSCC-Q, and JCM across different MCS indices under a slow Rician fading channel with $K_{\text{factor}} = 6$. The evaluation is performed on the Kodak dataset using PSNR, MS-SSIM, and LPIPS as metrics.

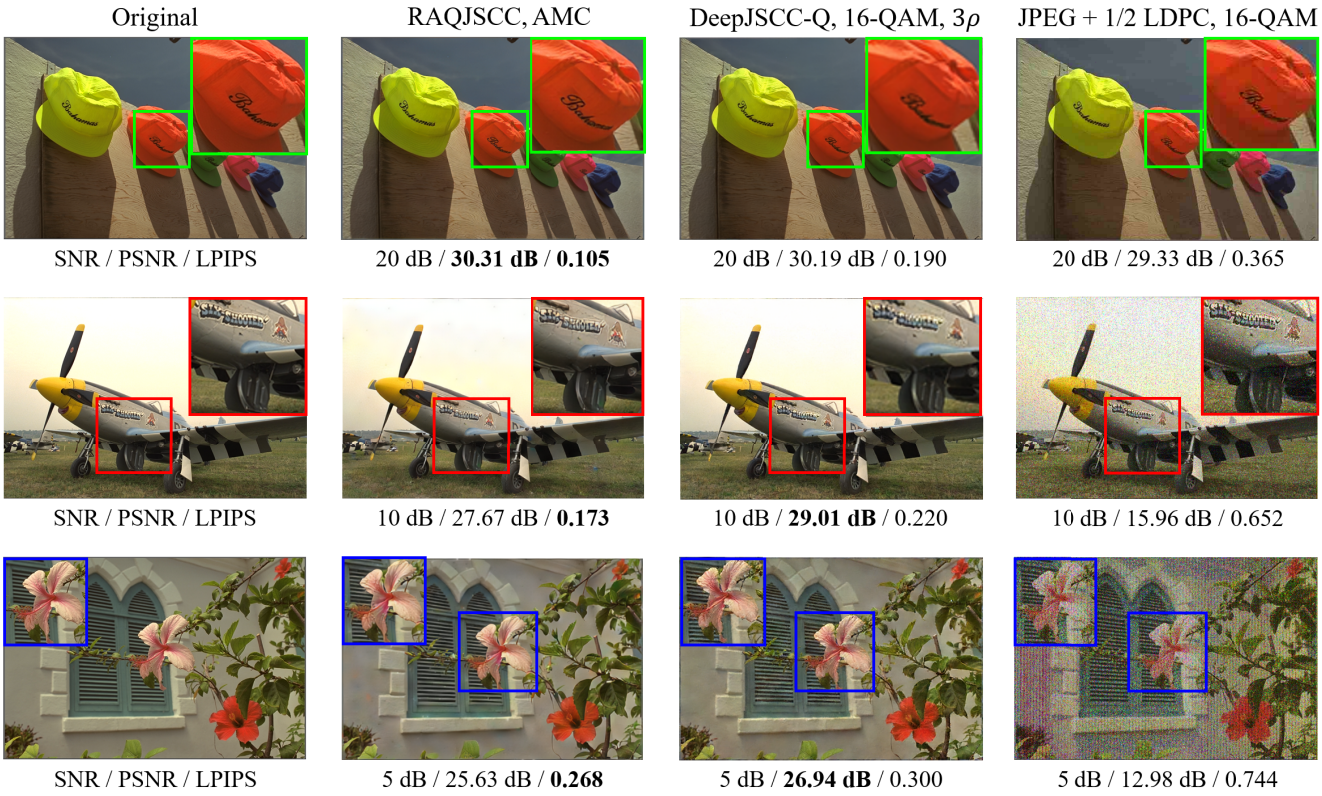
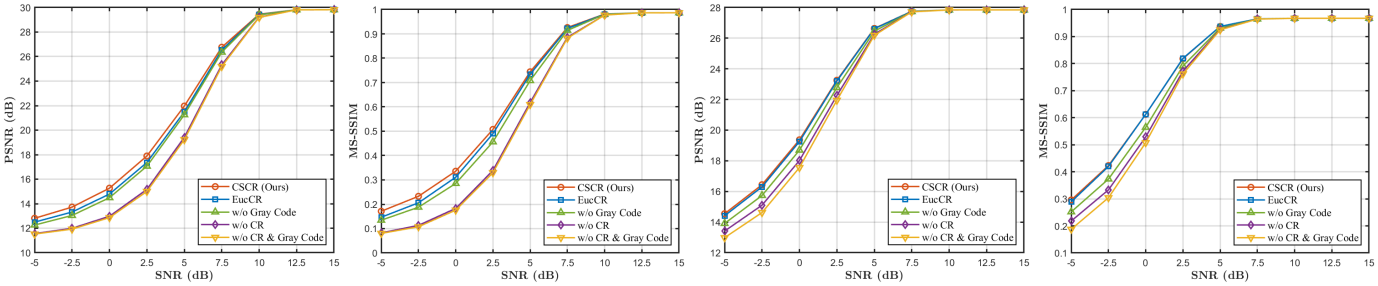


Fig. 6. Visual examples of image reconstruction from RAQJSCC, DeepJSCC-Q, and JPEG+LDPC on the Kodak dataset at different SNR levels (20 dB, 10 dB, 5 dB).

help the system better adapt its operating parameters to the current channel quality. As the receiver demands higher quality of service (QoS) levels, such adjustments would ensure that the MCS selection remains aligned with the actual channel conditions, resulting in better reconstruction quality and more reliable performance.

To further examine the generalization capability of the proposed system under more realistic wireless environments, we additionally evaluate the models over a slow Rician fading channel with a fixed $K_{\text{factor}} = 6$. It is noted that none of the compared models are explicitly trained to handle Rician fading. As shown in Fig. 5, RAQJSCC consistently achieves higher robustness across all SNR regions by employing the low-order MCS #0 (BPSK) configuration under harsh channel conditions, which aligns with the robustness characteristics typically observed in conventional digital communication sys-

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(a) PSNR, QPSK, 5ρ (b) MS-SSIM, QPSK, 5ρ (c) PSNR, BPSK, 6ρ (d) MS-SSIM, BPSK, 6ρ

Fig. 7. Evaluation of various codebook reordering strategies in RAQJSCC under AWGN channels. We compare five methods: the proposed CSCR with Gray code (*CSCR*), Euclidean-based CR with Gray code (*EucCR*) [26], CSCR without Gray code (*w/o Gray code*), Gray code mapping without CR (*w/o CR*), and no reordering (*w/o CR & Gray code*).

tems. In contrast, DeepJSCC-Q and JCM achieve their best performance at MCS #2, where higher modulation orders and BCRs yield improved efficiency. The overall trends in MS-SSIM and LPIPS are consistent with the AWGN results, demonstrating that the AMC capability of RAQJSCC effectively enhances noise robustness under fading channels.

Fig. 6 shows visual examples of reconstructed images on the Kodak dataset at different SNR levels. RAQJSCC consistently delivers high visual fidelity across all conditions. At 20 dB, where MCS #4 is selected, RAQJSCC produces an image that closely resembles the original, while 'DeepJSCC-Q (16-QAM, 3ρ)' and 'JPEG+LDPC (16-QAM, 3ρ)' exhibit blurring and character smearing. As the channel condition degrades, RAQJSCC adapts its transmission strategy via AMC, selecting lower MCS levels, such as MCS #2 at 10 dB and MCS #0 at 5 dB, to maintain robustness. Even under these challenging conditions, RAQJSCC preserves key structures and colors better than the baselines, demonstrating the benefits of its integrated AMC control and perceptual optimization. Note that while all quantitative comparisons in Section V employ the BPG+LDPC baseline, the qualitative results here retain JPEG+LDPC as the digital reference, since BPG decoding frequently failed below 10 dB due to bitstream corruption, making consistent visual illustration infeasible.

Overall, these results demonstrate that RAQJSCC operates as a structural adaptation mechanism within a single JSCC model, dynamically adjusting the effective MCS rather than relying on any form of model switching. This built-in flexibility enables RAQJSCC to support multiple code rates and modulation schemes in one unified framework, delivering robust and perceptually consistent reconstructions across a wide range of channel conditions. This capability allows it to outperform fixed-rate digital JSCC and separation-based coding schemes, ensuring a more stable quality of service as channel conditions vary. Moreover, fine-tuning the MCS index mapping or introducing additional intermediate MCS levels could further improve its adaptability and performance, particularly for intermediate SNR ranges where careful tuning of parameters is essential.

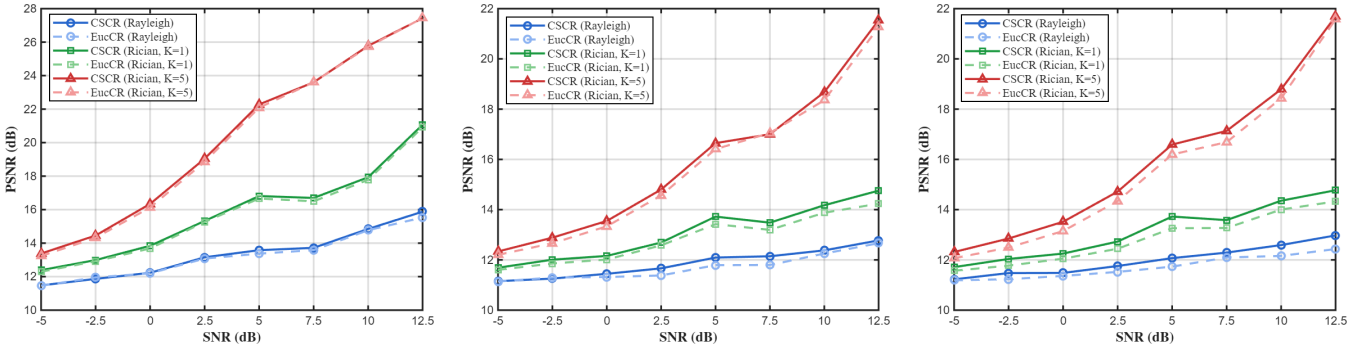
E. Impact of CSCR

We conducted an ablation study comparing five codebook index assignment strategies within the RAQJSCC framework under AWGN channels to evaluate the robustness provided by CSCR against channel errors. Experiments were carried out in two modulation scenarios: QPSK with BCR = 5ρ, and BPSK with BCR = 6ρ. In Fig. 7, we focus on the reconstructed image quality in terms of PSNR and MS-SSIM across a range of SNR values. The five methods compared are: (i) the proposed cosine similarity-based codebook reordering with Gray code mapping (*CSCR*), which uses our angular similarity metric to reorder codebook entries and then applies Gray codes for index assignment; (ii) Euclidean distance-based codebook reordering with Gray code mapping (*EucCR*) [26]; (iii) CSCR *without* Gray code mapping (*w/o Gray code*); (iv) Gray code mapping *without* any codebook reordering (*w/o CR*), which keeps the base codebook order but labels the codewords with Gray-coded indices; and (v) a baseline with neither reordering nor Gray code mapping (*w/o CR & Gray code*), where the codebook is used in its original form with conventional binary indexing.

As shown in Fig. 7, our CSCR consistently achieves the highest PSNR and MS-SSIM across most SNR regimes in both the QPSK and BPSK experiments. By effectively mitigating the impact of channel bit errors, CSCR delivers notably better image quality, particularly in low-SNR regimes. In the QPSK case (BCR = 5ρ), CSCR outperforms EucCR at nearly all SNR values, highlighting the benefit of using cosine similarity over Euclidean distance for codebook reordering. This advantage is most pronounced in the low-SNR regime: for example, at a very low SNR, CSCR achieves roughly 0.3 dB higher PSNR than EucCR and maintains higher MS-SSIM as well. Gray code mapping yields only a minor additional gain in the QPSK scenario. The performance gap between CSCR and its variant without Gray code is relatively small (on the order of 0.5 dB in PSNR at the lowest SNR). This suggests that most of the robustness improvement in QPSK comes from the cosine similarity-based reordering itself, while the Gray code contributes a smaller refinement in this setting.

In the BPSK experiment (BCR = 6ρ), we observe that CSCR still provides the best performance, but the gap between CSCR and EucCR is less pronounced than in the QPSK case. Their PSNR and MS-SSIM curves are quite close at moderate and high SNRs. These results indicate that, under BPSK modulation, choosing between cosine similarity and Euclidean reordering is slightly less critical when the channel conditions are good. However, Gray code mapping has a

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(a) PSNR, BPSK, 5ρ (b) PSNR, QPSK, 3ρ (c) PSNR, QPSK, 5ρ

Fig. 8. Evaluation of CSCR and EucCR under different fading channels. Results are shown for Rayleigh and Rician fading channels, where a larger Rician K-factor corresponds to stronger LoS conditions.

more noticeable impact in the BPSK scenario. At low SNR, CSCR (with Gray code) achieves about 0.6 dB higher PSNR than CSCR without Gray code, and similarly yields a higher MS-SSIM, demonstrating that Gray coding provides a significant boost in error resilience when only one bit is transmitted per symbol. Even the method that uses Gray code without any CR outperforms the baseline without Gray code, especially in the low-SNR regime, underscoring Gray code's effectiveness in mitigating bit errors. Intuitively, this is because Gray coding ensures that adjacent index values differ by only a single bit, so a one-bit channel error is less likely to produce a large distortion in the decoded image. By combining Gray code mapping with our cosine-based reordering, CSCR achieves the best of both strategies, yielding the highest robustness among all tested methods.

To further investigate the generalization of the CSCR strategy under more realistic fading environments, we additionally evaluate its performance across Rayleigh and Rician channels, as shown in Fig. 8. For the Rician model, we consider two representative LoS conditions with K_{factor} values of 1 and 5, where a larger K_{factor} corresponds to stronger LoS dominance. In the BPSK case with BCR = 5ρ, the overall PSNR difference between CSCR and EucCR remains small (within 0.1–0.3 dB), yet CSCR consistently achieves slightly higher reconstruction quality across all channel types. In the QPSK experiments, however, a more noticeable margin is observed: CSCR provides up to a 0.5-dB PSNR improvement over EucCR, particularly under Rayleigh fading and Rician ($K_{\text{factor}} = 1$) conditions where multipath-induced index perturbations are more frequent. This consistent advantage across distinct fading statistics suggests that cosine similarity-based reordering provides enhanced robustness not only in additive noise but also in the presence of channel amplitude and phase variations.

We further quantify the computational overhead of the Codebook Adaptation Module and the subsequent CSCR on an NVIDIA RTX 4090 GPU to validate their efficiency in real-time settings. The codebook adaptation module requires approximately 9.7 ms to generate a codebook with $\widetilde{K} = 64$ and up to 143.9 ms to generate one with $\widetilde{K} = 1024$. The subsequent CSCR process adds approximately 6.9 ms of latency at $\widetilde{K} = 64$ and up to 138.6 ms at $\widetilde{K} = 1024$. Since both operations are performed only once per MCS configuration and the resulting codebooks are cached and reused during transmission, their combined one-time latency is negligible for real-time operation. These results confirm that the adaptation and reordering modules together introduce marginal computational overhead while enabling the robustness benefits of CSCR.

Overall, these results confirm that cosine similarity-based reordering is a highly effective strategy for enhancing the error robustness of VQ-based JSCC. The CSCR approach consistently surpasses the Euclidean distance-based reordering approach (EucCR) derived from prior work [26], which indicates that capturing the angular relationships between high-dimensional codebook embeddings is more beneficial than using Euclidean distance. In high-dimensional spaces, Euclidean distances can become less informative, a manifestation of the curse of dimensionality, whereas cosine similarity focuses on the direction of vectors, aligning more closely with semantic similarity in the embedding space. Moreover, the incorporation of Gray code indexing further improves reliability by reducing the impact of bit flips on the transmitted codewords. While the Gray code provided only a modest gain under QPSK, it became crucial for improved performance under BPSK, illustrating that its relative benefit can depend on the modulation and bit rate. By jointly leveraging cosine similarity-based reordering and Gray code mapping, our RAQJSCC system achieves superior reconstruction quality over a wide range of channel conditions, maintaining higher PSNR (often by more than 1 dB at low SNR) and MS-SSIM than those achieved with Euclidean-based reordering or no codebook reordering. This demonstrates the practical promise of the proposed CSCR method for wireless image transmission, as it significantly bolsters the system's robustness and adaptability in the face of unpredictable channel noise.

VI. Conclusion

This paper presented RAQJSCC, a unified digital DeepJSCC framework that enables rate-adaptive transmission by directly controlling the codebook size within a VQ-based discrete representation. Unlike existing approaches that rely on modifying the quantization structure or introducing external rate-control mechanisms, the proposed framework achieves rate adaptation while preserving a consistent encoder–decoder mapping. By integrating RAQ with a digital transmission pipeline,

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we extend rate-adaptive quantization beyond compression-oriented settings to physical-layer JSCC systems. In this context, we showed that the codebook size serves as a direct control variable for the transmission rate, providing a simple yet effective mechanism for multi-rate operation within a single model. To address the sensitivity of VQ-based transmission to index errors, we introduced CSCR, which aligns codeword ordering with semantic similarity. This design can be interpreted as a source-channel interface that reduces the distortion induced by bit-level perturbations, rather than merely improving robustness at the signal level. Furthermore, by incorporating an AMC control unit, we connected the learned representation with practical link adaptation mechanisms. Our analysis showed that, in digital JSCC systems, modulation order and codebook size jointly determine both reliability and distortion, and that conventional MCS design does not always align with end-to-end reconstruction objectives. Extensive experiments demonstrated that RAQJSCC achieves consistent performance gains across a wide range of channel conditions, while avoiding the need for multiple models. Beyond performance improvements, our results suggest that effective digital JSCC design should explicitly account for the interaction between discrete representations, modulation schemes, and channel-induced perturbations.

Beyond the core design, the proposed framework is model-agnostic and can be integrated with more advanced VQ-based architectures, including recent generative models, without requiring architectural modifications. Future work will extend the current framework toward more realistic wireless environments, including fast time-varying and frequency-selective channels, and explore distortion-aware adaptation strategies such as learning-based or reinforcement learning-based control. In addition, extending the framework to other modalities, such as speech and multimodal semantic communication, is a promising direction.

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